

Chicken Nutrition

A guide for nutritionists and
poultry professionals

By Rick Kleyn



CONTEXT

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Preface

As a practising commercial poultry nutritionist I am often called upon to share my knowledge with others. The latter includes groups of poultry producers, fellow practitioners and university students. To this end I have been running a 3-day introductory course in poultry nutrition for over a decade, which has been attended by hundreds of people from many countries. This book began as study material for that course. The fast pace of change in the poultry industry has necessitated continuous revision and this volume represents the culmination of years of work. Those of us who work in the poultry industry will know that production systems and management practices have an impact on the way in which chickens should be fed, how production results should be interpreted and how any associated

problems should be solved. Where relevant, aspects of production systems that are relevant to nutritional decision-making are discussed.

Feed makes up about 70% of the production cost of any poultry production system. While many poultry producers have an excellent understanding of poultry pathology few understand what it is that the nutritionist is trying to achieve. This is borne out by the comment of one of our course delegates, who thought, 'amino acids were something that one put into the swimming pool'. He has assured me that he now knows better! It is vital that all poultry practitioners have at least a working knowledge of nutrition, even if it is only to keep feed suppliers on their toes.

To many people poultry nutrition appears to be more complex than the nutrition of other species. Scott (1991) makes the point that more is known about poultry nutrition than any other species, which is partly true because of the practical incentive to produce the most highly efficient chicken feeds. It is, however, also true because the chicken is one of the most suitable animals for studying basic nutrition. There is, therefore, a large knowledge base for poultry nutrition and, in many ways, this simplifies poultry nutrition rather than complicates it.

It needs to be borne in mind that there are far greater similarities between species (including man) than there are differences between them. Not only do we share much of our genetic material (DNA) with other animals but we also

share all of the critical biochemical pathways that are involved with nutrition. This means that birds of different species (genotype) may have very similar nutrient requirements. For example, the requirements of a 1 kg broiler are very similar to those of a 1 kg Peking duck. However, birds of a similar genotype but that are phenotypically different; for example, a 100 g broiler chicken will have very different nutrient requirements to a broiler weighing 1.5 kg.

An aspect that makes poultry nutrition different to human, or possibly even ruminant nutrition, is that poultry flocks represent a population of individuals, each with a different production potential, feed intake and metabolic efficiency. When feeding poultry the aim should be to maximise the profitability of the flock as a whole. If this aim is not met the high performing individuals may well be underfed and not able to perform to their optimum potential.

This book explores some of the underlying theory of poultry nutrition rather than providing a step-by-step methodology for formulating poultry diets. If the underlying principles are understood, it makes it easier to recognise what is seen in the field and to be equipped to solve any problems. Teaching nutrition is always a little difficult. It is a little akin to a bicycle wheel. Each spoke needs to be dealt with individually, and it is not until all of the spokes are in place that the wheel of our understanding is complete. Despite this shortcoming it is hoped that the book will serve as a useful resource for people involved in poultry production, feed manufacture and to students who are trying to grasp the intricacies of poultry nutrition.

The most important aspects of the diet as far as commercial nutritionists are concerned are those that make up the largest proportion of the costs. These are energy, protein and a few of the macro-minerals. This book will, therefore, concentrate on these aspects of nutrition, rather than discussing the nutrition of trace mineral and vitamins. It should be borne in mind that as the genetic potential of poultry increases some of the vitamin and trace minerals may begin to limit production. Thus the importance of vitamin and mineral nutrition is likely to increase in the future.

It needs to be remembered that nutrition is a scientific activity and should, therefore, follow a traditional scientific procedure or methodology. Science is a logical, objective process for testing ideas, and reaching a conclusion; the scientific data need to be interpreted so that meaningful decisions can be made. For this reason, a chapter on the scientific process is included.

The objective of this book is to teach you the golden rules of poultry nutrition. These are:

- 1 There are no free lunches!
- 2 Never evaluate a feed in terms of what it includes; rather judge a feed's level of fitness by what has been left out of it.
- 3 What we do may not necessarily be correct but we must ensure that we are wrong consistently.
- 4 There is no black magic involved in nutrition (it is a science).

No work of this nature can happen in isolation. I would like to acknowledge the help of past and present colleagues at SPESFEED, namely Brett Roosendaal, Helena Smuts, Christel Coetzee, Colleen Engelbrecht and Bianca Losper who have made significant contributions to the information contained in the book. A special word of thanks goes to Steve Leeson who is always happy to share his knowledge and has graciously allowed me to use much of his material. Peter and Natalie Chrystal have been contributors and constructive critics of the book over the years. I have worked with many specialist veterinarians and would like to thank them, especially Chris Henderson and Herman Bosman, for the help and insight that they have given me. Anne Marangos was responsible for editing my English and my thanks go to her for her thoroughness. Wes Ewing and his team at Context have been enthusiastic and innovative throughout the publishing process. Andre Pretorius kindly took most of the photographs used in the book. Lastly, I would like to pay tribute to all of my fellow practitioners who wittingly or unwittingly have contributed to my knowledge of poultry nutrition. If for any reason, through oversight or ignorance, an acknowledgement has been omitted it is regrettable. This will immediately be rectified in any future editions of the book, be they electronic or traditional, once drawn to our attention

Rick Kleyn

Foreword

This new publication by Rick Kleyn is a welcome addition to those intimately involved in poultry nutrition and feeding. It has been some time since there has been a balanced overview of modern concepts of basic nutrition together with practical application for on-farm feeding management. For some time Rick Kleyn has undertaken a training seminar in poultry nutrition for professionals in Southern Africa and beyond. This book represents material for this course that has evolved over 15 years of development. Rick Kleyn is a consulting nutritionist, most active in Africa but also

spends time in most countries in Europe and so his wealth of practical world-wide knowledge is reflected in this book. There are 18 main Chapters starting with all the main nutrient classes and then application of this information in separate Chapters on broiler, broiler breeder and layer nutrition. The unique attribute of the book is found in the following 7 chapters that detail such current important topics as ingredient evaluation, enzyme use, nutrition and health and quality assurance as well as an interesting unique critique of the "scientific process". Being a practicing nutritionist the author provides

very interesting inside information on the hands-on process of feed formulation which to my knowledge is missing in most other texts on poultry nutrition. Overall, a very readable and informative text on all aspects of poultry feeding and feed program development. The text will be of use not only to commercial nutritionists but also to students involved in all aspects of poultry production, and most other professionals in the various poultry industries.

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Nutrition and nutrients



Fact
Essential nutrients cannot be synthesised by the bird and must be supplied in the diet.

The science of nutrition can be defined as 'the process of providing a population of animals with a diet that allows for effective function of the metabolic pathways required for growth, maintenance, work, immunity and reproduction'. Nutrition encompasses the procurement, ingestion, digestion and absorption of the chemical elements that serve as food. In addition, it includes the transport of these elements to all regions within the animal organism in the physical and chemical forms most suitable for assimilation and use by the cells. The primary concern of any nutritionist is to ensure that the diets being offered to an animal contain sufficient quantities of each nutrient in order for it to function, produce and reproduce normally. Nutrition is a quantitative science, requiring not only an accurate description of the molecular details of digestion, metabolism and excretion but also an accurate estimation of their rates. Most importantly, nutrition is an economic science in that whatever is done in nutritional terms may have a direct bearing on the profit and loss of the poultry operation.

The monogastric digestive system operates in the stomach and small intestine, which is well adapted to dealing with lipids, sugars, starches and proteins (see Chapter 7). The highly efficient caecal and large intestinal digestive system operates through the help of a substantial bacterial population, which will deal with nutrients in plant tissues high in non-starch polysaccharides. Birds can readily discriminate the different nutritive values of a wide variety of feedstuffs, which enables self-choice feeding and the automatic balancing of the daily diet.

The Oxford dictionary defines a nutrient as 'any substance, which provides for, or contributes towards, nourishment'. These substances are mostly of organic origin (plants and animals) but may also be minerals. Increasingly, nutrients (vitamins and amino acids) are being

manufactured by chemical processes and these synthetic nutrients are playing an ever-increasing role in nutrition. Nutrients are components of the diet that have specific functions within the body and contribute to tissue maintenance, growth and health of the animal. **Essential nutrients** are those components required by the animal that cannot be synthesised in sufficient quantities to meet the animal's requirement. It is, therefore, essential that they be supplied in the diet. **Non-essential nutrients** on the other hand can be synthesised by the body in sufficient quantities. Fine Along with energy, all animals have requirements for each of the 6 classes of nutrient, namely, water, carbohydrate, fat (lipid), protein, vitamins and minerals.

Increasingly, there is talk of what are being called nutricines. These are defined as molecules that are not traditional nutrients yet have important biochemical functions. Examples would be organic acids and enzymes. In the past, we would have called these non-nutritional additives. Some authors refer to nutrients such as essential fatty acids as nutricines, but this is not strictly correct.

In its broadest sense, nutrition is governed by the laws of physics, the most important of which is the First Law of Thermodynamics, which states that matter and energy are always conserved. They cannot be created or lost. Stated differently, everything the animal consumes is accounted for: it is either undigested (lost via the faeces) or it is absorbed and used by the body, with byproducts being excreted in the urine, faeces and through respiration or lost as heat. It has already been mentioned that all fauna share much of the same DNA and, more importantly, the complex biochemical pathways that constitute metabolism. These are unlikely to change through genetic selection, so the likelihood that we will be able to change the way in

which chickens utilise nutrients and energy at the cellular level probably does not exist.

Michael Pollan (2008) believes that we have entered the age of 'nutritionism'. He defines it in the following way: Nutritionism is not the same as nutrition. As the -ism suggests, it is not a scientific subject but an ideology. Ideologies are ways of organising large swathes of life and experiences under a set of shared but unexamined assumptions. A reigning ideology is a little like the weather – all pervasive and virtually impossible to escape.

A classic example of nutritionism was the widespread belief that cholesterol in eggs caused an increase in heart disease – now something known to be untrue. Another example would be the belief that the protein level of a diet or ingredient is its most important attribute, or that its fat content is its least desirable attribute. Sadly, marketing departments often do not understand the difference between nutrition and nutritionism.

Nutrition is not complicated. In its simplest terms the feed we offer our birds contains only a few elements. These are:

- Water.
- Energy to fuel all life processes.
- Nutrients (protein, fat, vitamins and mineral), which are the building blocks of all tissue.
- Non-nutritive additives (medication, enzymes and colourants).

Similarly, the bird's needs are simple. They eat their food for:

- Maintenance – staying alive, largely determined by body size.
- Growth, which includes the skeleton, lean tissue, fat tissue and feathers.

Body metabolites



Fact

The Krebs cycle yields adenosine triphosphate (ATP) which represents 'free' energy in the animal.

Importantly, nutrients and metabolites are regarded as being present in several pools in the body, although no nutrient exists in a single homogenous pool (Figure 1.3). At the simplest level, there are three pools, although more complex systems do exist. The **functional pool** is directly involved in one or more bodily functions, while the **storage pool** provides a buffer of material that can be made available for the functional pool. A **precursor pool** provides a substrate from which the nutrient or metabolite can be synthesised. Importantly, the essential nutrients do not have a precursor pool (Macdonald *et al.*, 2011).

The starting points of metabolism are the substances produced by the digestion of food. For all practical purposes, we may regard the end products of carbohydrate digestion in the monogastric animal as glucose, with very small amounts of galactose and fructose. These are absorbed into the portal blood and carried to the liver. In ruminant animals the major part of the dietary carbohydrate is broken down in the rumen to acetic, propionic and butyric acids (volatile fatty acids). Glucose joins the liver glucose pool from where it is converted partly into glycogen and stored, or into a glycerophosphate used for triglyceride synthesis. The remainder of the glucose enters the systemic blood supply

and is carried to various body tissues where it may be used as an energy source, as a source of co-enzymes in fatty acid synthesis or for glycogen synthesis.

Digestion of proteins results in the production of amino acids that are absorbed by the intestinal villi into the portal blood. The amino acids are then carried to the liver where they join the amino-acid pool. They may then be used for protein synthesis. They may also pass into the systemic blood and join the amino acids that are produced as a result of tissue catabolism, to provide the raw material for synthesis of proteins and other biologically important nitrogenous

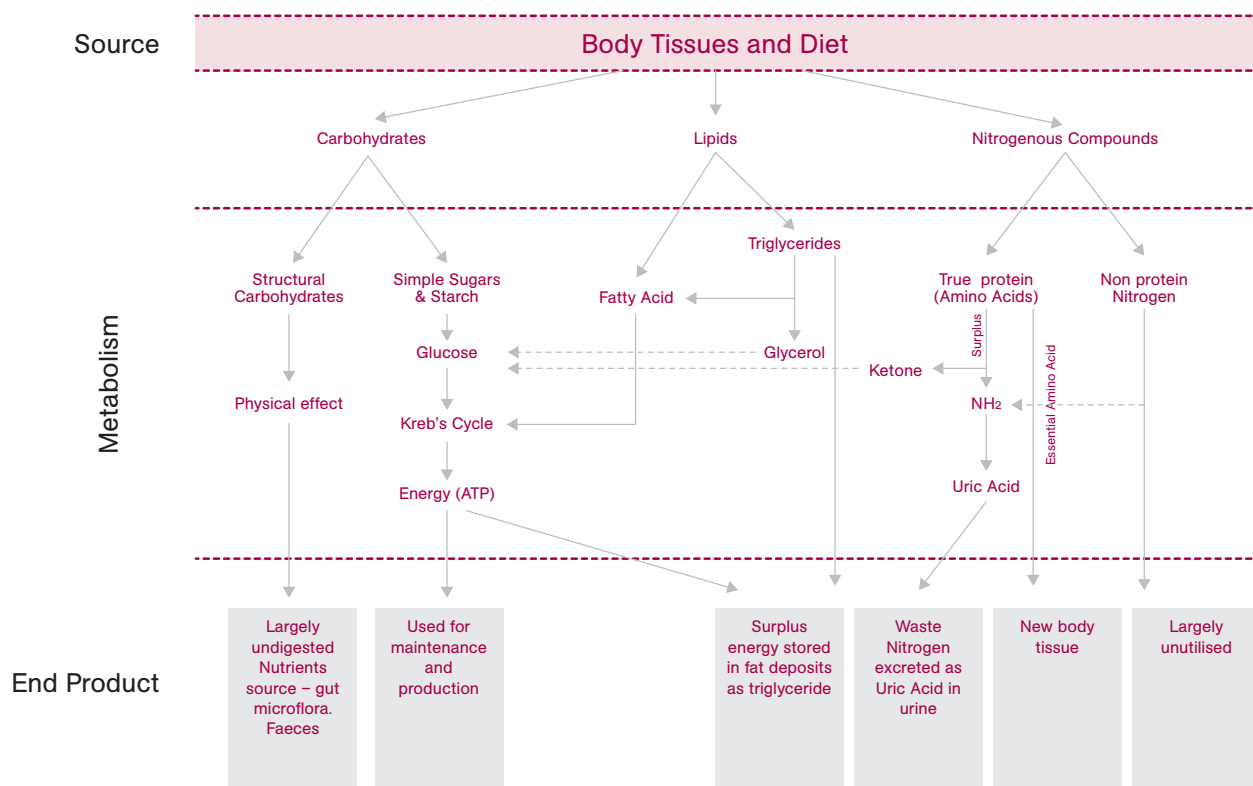


Figure 1.4 Sources and fates of major body metabolites (redrawn from MacDonald *et al.*, 2011)

Carbohydrates in feed ingredients



Fact
Many carbohydrates are either poorly utilised, or not utilised at all by monogastric animals.

Table 3.3 The various carbohydrates found in feed ingredients

	Name			Principle sugars	Forms/names		Principle enzyme	Comments
Class	Sugar	NSP	Sol.					
Mono	+		++	Glucose	Glucose			Primary energy substrate.
Mono	+		+	Galactose	Galactose			Converted to glucose. Not as efficient as glucose.
Mono	+		-	Fructose	Fructose			No transport system in poultry.
Mono	+		-	Lactose	Lactose		Lactase	Milk sugar. Lactase enzyme absent in poultry so can't utilise. Is added to diets as a pre-biotic.
Di	+		++	Sucrose	Glucose + fructose		Sucrase	Sugar in the kitchen. In most plants sucrose is converted to starch.
Di	+		+++	Maltose	Glucose + glucose			Does not occur naturally. Breakdown product of starch.
Di	+		--	Raffinose	Galactose + fructose			High levels in soya beans. Neither hydrolysed nor absorbed by monogastrics.
Poly	+		++	Starch	Glucose	Amylose (Helix)	Amylase	Water soluble though not readily digestible. Comprises 25-25% of maize starch.
Poly	+		+++		Glucose	Amylopectin (Branched chains)	Saccharase	Not soluble. Absorbs water and swells. Cooking leads to gelatinisation.
Poly	+			Pectin	Galactose chains.			Primary cell wall of plant tissue. Side branches.
Poly		+	+	Cellulose				
Poly		+	+	Hemi-cellulose		Xylans	Xylanase	Major constituent of seed coats of cereal grains. Wheat bran contains 25%.
Poly		+	-			Mannans		No transport system in poultry.
Poly		+	-			Glucomannans		
Poly		+	+	β-glucans		Glucose with β 1-3 linkage	Glucanase	At boundary between hemicelluloses and gums. Best known is oat glucan. Very viscous in solution.
Poly		+	--	Gums				High viscous at low temperature.

Notes: Classes are monosaccharides, disaccharides and polysaccharides. NSP = non-starch polysaccharides.
Sol. is an estimation of the solubility in the monogastric digestive system.

acids also occur naturally (Figure 3.10). Dietary lipids supply energy, essential fatty acids and pigments. Animals are able to synthesise saturated fatty acids but are unable to synthesise linoleic and α-linolenic acid. These two fatty acids are,

therefore, considered as **essential fatty acids** for animals. A classification of fats is shown in Figure 3.11.

The energy value of fats and oils depends mainly upon the absorption rate of the

fatty acids from the intestinal tract. Since fatty acids are not excreted in the urine, their metabolizable energy values are directly related to their absorption. When fats are included in diets for growing animals, the efficiency of utilisation of

Amino acid balance



Fact
The protein quality of a diet is determined by the level of the first limiting essential amino acid.

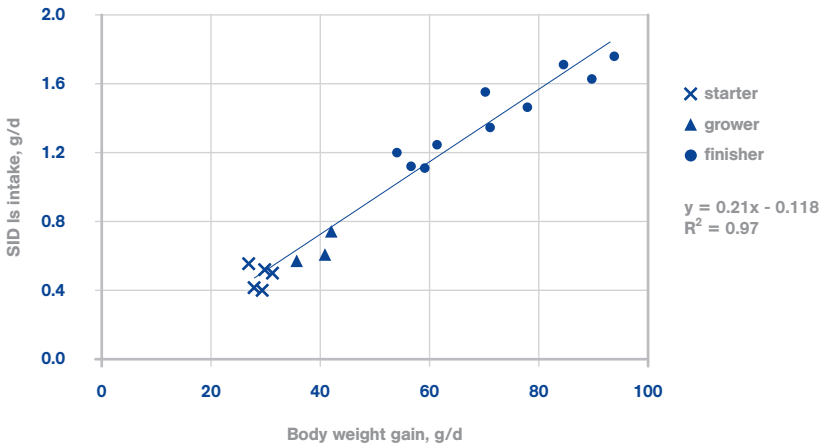


Figure 4.4 SID lysine intake (g/day) for optimal body weight gain (g/day) (literature review: 20 trials covering various periods from 0–49 days of age) (Relandeau and le Bellego, 2004)

is not ideal for growth. By balancing the diet for lysine (the first limiting amino acid) and ensuring that all other amino acids are correct as determined by the ideal amino acid profile, a 'near perfect' diet is created. This will have a high biological value, a consequence of which is that the

amino acids in the feed will be converted into lean meat mass with a high level of efficiency. Working on the Ideal protein basis will also allow for maximum lean meat production with a minimum intake of amino acids. Surprisingly, the commercial utilisation of protein is only about

55%, which is a result of poor protein digestibility, low biological value of the proteins being consumed and overfeeding of non-essential amino acids.

Amino acids and crude protein

Theoretically, protein should not affect a bird's performance as long as all the amino acids are at required level and there is no overt antagonism. Poorly balanced diets result in inferior performance, suggesting the utilisation of the first limiting amino acid is impaired through a reduction in its utilisation rather than its absorption. Boorman and Ellis (1996) suggest that a more likely explanation for reduced performance on when diets with an amino acid imbalance are fed is through a reduction in the net energy of the diet resulting from an increase in gluconeogenesis. Morris and Gous (1999) show that disappointing results may be a consequence of the requirement for the limiting amino acid increasing with increments of crude protein. This effect can only be explained as some form of generalised amino acid imbalance: these authors were able to predict these relationships (Table 4.4). The results indicate that any deficiencies in a diet that are created by using a poor quality protein source cannot be rectified by simply providing more of the ingredient. Although confusing, in the case of laying hens the same responses are not measured and it would appear that it is only the levels of limiting essential amino acid that are of relevance.

Table 4.4 Optimum ratio of amino acid to protein (Morris and Gous, 1999)

% Lysine = 0.057% CP
% tryptophan = 0.012% CP
% Methionine = 0.025% CP

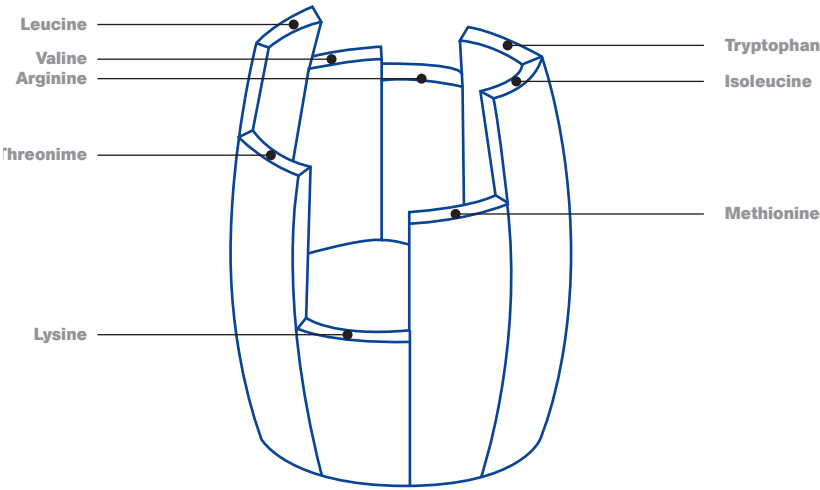


Figure 4.5 The barrel analogy for amino acid balance

Organic trace minerals



Fact
Expected dietary levels should not be confused with either requirements or maximum permissible levels.

of Se (selenium salinate) are very poorly available. In addition to this, the material is highly toxic. Organic forms of Se would appear to be more digestible and are increasingly being included in animal diets. A Se deficiency causes a condition in chicks known as exudative diathesis. This is normally seen at $\pm$ 6 to 12 weeks of age, and symptoms are oedema, in the form of weeping of the skin in the inner surface of the thighs and wings. Birds tend to bruise extremely easily. It is of interest to note that broilers fed diets containing low levels of Se have higher drip loss when slaughtered. The pancreas also becomes fibrotic, which results in reduced production of lipase, trypsinogen and chymotrypsinogen. This pancreatic involvement also results in reduced egg production and feed conversion in adult birds. High levels of Se also reduce hatchability in adult birds.

Cobalt

Cobalt (Co) is sometimes included in premixes. However, it is a presumed to be a human carcinogen when inhaled. Co is a precursor to vitamin B₁₂, and when adequate levels of vitamin B12 are provided in the diet, Co supplementation is not required.

Mineral toxicosis

The storage capacity of the various microminerals varies considerably. Until storage sites are saturated with microminerals, plasma levels do not increase proportionally with uptake. Once storage is saturated, the concentrations of circulating microminerals increases rapidly until toxic symptoms occur. Toxicity is defined as the stage at which production declines and the extent and severity of symptoms observed are a function of previous micromineral nutritional intake as well as the duration of excessive intake. The mineral levels of the feed ingredients can vary due to soil factors and the quantity and availability of the minerals in the soil. Mg and Se, for example, may

occur in crops at levels that they may have adverse effects on animals. Natural water supplies may contain excessive levels of S, Na, Mg and Fe and there may even be some industrial contaminants in the water. This, coupled with the problem of cross-contamination in the premix plant and feed mills, can give rise to higher than expected levels of minerals occurring in the feed. When the diet is contaminated with the heavy metals, widespread feed refusal will be the result.

Vets are very aware of this problem and analyse feed to check the mineral levels in the diet. Where problems often arise is in the interpretation of the results. Many people understand the published *minimum* levels of minerals to mean the 'expected' level in the diet. This is not the case. Rather, these are the levels that need to be exceeded in order prevent deficiency symptoms. Animals are capable of selectively absorbing minerals and, normally, the higher the levels contained in the diet, the higher the levels in the faeces. The 1980 NRC publication entitled *Mineral Tolerance of Domestic Animals* the term *maximum tolerable level* is defined as the dietary level that, when fed for a limited period, will not impair animal performance and should not produce unsafe residues in human food derived from the animal. Greater sensitivity to high mineral levels can be expected in young, pregnant, lactating malnourished or diseased animals. It is in keeping with good nutritional practice to maintain mineral intake at required levels, which are generally well below maximum tolerable levels. Table 6.8 gives some indication as to what these figures should be.

Organic trace minerals

The so-called organic minerals (this is a misnomer because all minerals are by definition inorganic) comprise minerals bound to organic compounds such as protein or carbohydrates. Metals bound

Table 6.8 Minimum and maximum tolerable levels of minerals in poultry diets (ppm) (NRC, 1980)

Mineral	Minimum requirement	Maximum tolerable level
Cobalt	-	10
Copper	8	300
Iodine	0.35	300
Iron	80	1000
Magnesium	-	3000
Manganese	60	2000
Selenium	0.15	2
Zinc		1000

to organic ligands by coordinated bonds can dissociate within the animal's metabolism whereas the covalent bonds of inorganic salt cannot. Where the ligands are small molecules, such as amino acids, digestibility is higher than for peptides or proteins, these minerals are better utilised by animals. After absorption, organic minerals may present physiological effects, which improve specific metabolic responses such as the immune response. Many studies have demonstrated the benefits of metal-amino chelates on animal metabolism but the detection of positive effects on live performance is less consistent (Vieira, 2008).

There are a number of categories of organic trace minerals, with the metal amino acid complexes probably being the most common. They result from combining a specific soluble metal salt (such as zinc, copper, and manganese) with an amino acid. A specific amino acid may also be used. Metal proteinates result from the chelation of a soluble salt with amino acids and/or partially

Key pointers



Fact

All inorganic minerals need to be ionised before they can be absorbed.



KEY POINTERS

01

The large insoluble molecules that make up food need to be degraded into simple molecular compounds before they can cross the mucosa of the intestine and enter general circulation.

02

Animals ingest simple molecules and not specific ingredients.

03

Digestion involves a combination of mechanical, chemical and microbial activities that contribute to the degradation of feed ingredients.

04

Development of a functional gizzard is important as this serves as a pacemaker for the GIT.

05

Absorption involves the transport of food molecules across the mucosa of the intestine and into the circulatory system.

06

Most absorption takes place in the small intestine, while the caeca (large intestine) only plays a small role.

07

Animals are only able to absorb molecules through the mucosa of the intestine – and not via the skin as many cosmetic companies would have us believe.

08

Digestion is under both voluntary and involuntary control.

09

Unlike mammals, reflux forms an important part of avian nutrition, allowing for the more efficient digestion and absorption.

Withdrawal diets



Fact
Carcass yield is reduced when feeding whole grain.

Table 10.14 Response of male broilers to feeding cracked maize from 7 days of age when starter cost \$209/ton and maize \$117/tonne) (Leeson *et al.*, 2003)

Treatment	49 day weight (g)	Total feed (g/bird)	FCR	Carcass weight (g)	Breast (% of carcass)	Protein intake (g/kg)	Feed cost (c/kg bodyweight)
Control	3030	5771	1.93	2234 ^a	21.5 ^a	371	52.4
Start+Wheat	2922	5722	1.99	2125 ^b	19.6 ^b	364	48.2
Start+Maize	2917	5693	1.98	2130 ^b	20.0 ^b	340	47.2

Table 10.15 Proportion of intake of starter and maize

Age	Starter (%)	Maize (%)
0–7 days	100	0
7–21 days	87.3	12.7
21–35 days	65.7	34.3
35–49 days	57.1	42.9

- Birds exposed to high temperatures (33°C) day and moderate (20°C) night temperatures dramatically reduce their feed intake during the hot time of the day, particularly the protein component of the diet. This increases when it gets cooler. This enables choice fed birds to grow faster and convert feed more efficiently.
- An energy saving is achieved by feeding the grain whole. First, the energy expenditure of grinding, steam addition and pelleting is eliminated and, secondly, the bird obtains more energy from coarse particles (Table 10.36). The gizzard needs to grind the whole grain and so the size of the gizzard increases significantly. The grain is retained for a longer period of time and more thoroughly exposed to the digestive enzymes of the proventriculus. There is evidence that the active

gizzard helps to reduce the oocyst excretion and increases the bird's immune mechanism.

- The carcass yield is reduced with choice feeding because the digestive tract increases in size and, therefore, forms a larger portion of the total live mass.
- Feed conversion becomes less efficient but feeding costs can be reduced to compensate.

Feeding whole maize

Dozier *et al.* (2010) looked at the benefits and costs associated with feeding different forms of feed to broilers. All diets were identical in terms of formulation and nutrient content. A standard starter crumble was fed for 2 weeks, at which point one of four experimental diets was offered. These were good quality pellets,

low quality pellets, good pellets to which 20 % and 30% of the formulated maize was added back to the grower and finisher diets post pelleting, and a mash produced with a roller mill. Even though the stocking density was only 14 birds/m², there were significant differences between treatments (Table 10.16). Interestingly, the results achieved with the mixture of rolled maize and pellets were the same as those achieved with good pellets alone. If the pelleting costs of around €5 per ton are considered, this may represent an opportunity for producers to not only save money but also to reduce the energy costs of pelleting.

Withdrawal/post finisher diets

It has been usual practice to remove the vitamin/mineral pack as well as the medication during the last 5 days of growth (Table 10.17). Many studies have

Table 10.16 Growth and performance of Ross 708 male broilers fed diets differing in feed form from 1–42 days of age (Dozier *et al.*, 2010)

	Revised PDI*	Weight (g)	FCR	Mortality (%)
Good pellets	85.5	3.167 ^a	1.712 ^{ab}	2.0
Poor pellets	52.5	3.093 ^{ab}	1.694 ^b	1.3
Added maize	87.0	3.141 ^a	1.715 ^a	1.6
Mash	-	3.053 ^b	1.695 ^b	1.5

Note:* PDI is the Pellet Durability Index.

Rickets



Fact

Rickets occurs in young birds and is characterised by soft bones and beak.

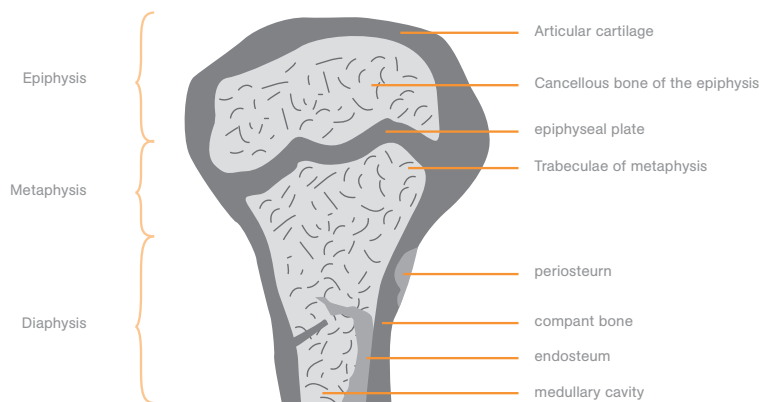


Figure 12.1 Schematic representation of juvenile long bone

growth plate, where the resultant disruption in nutrient supply means that the normal process of ossification does not occur. The exact cause of TD is unknown, although incidence can quickly be affected through genetic selection, the condition apparently being affected by a major sexlinked recessive gene. The primary breeding companies are aware of this and have done much to reduce the incidence of TD. Dietary electrolyte imbalance and particularly high levels of chloride seem to be a major contributor in many field outbreaks. More TD is also seen when the level of diet calcium is low relative to that of available phosphorus (Table 12.9), although the introduction of phytase may have changed this. Treatment involves adjustment of dietary levels of Ca:P and consideration of dietary electrolyte balance. Diet changes rarely result in complete recovery. TD can be prevented through reducing growth rate, so programmes of light or feed restriction must be considered in relation to economic consequences of reduced growth rate.

Chondrodystrophy

Chondrodystrophy is a general disorder of the growth plate in long bones

that eventually impairs linear, but not appositional, growth. The impairment of cartilage proliferation results in an impaired linear growth of bone though the bone width continues to increase. The condition was originally termed perosis (bowed and twisted legs). It is probably the major cause of leg problems in broiler chickens, affecting many birds to some degree. As many as 2–3% of males will be affected up to 40 days of age, while in older birds (42 days plus) a 1% incidence per week is not uncommon. Chondrodystrophy can simply be induced by feeding diets deficient in manganese or choline, although similar bone characteristics are also seen in birds deficient in zinc or most of the B vitamins. Chondrodystrophy still occurs in diets that are apparently well fortified, so nutrient bio-availability and/or antagonists (mycotoxins) are likely to be involved.

Femoral head necrosis

Femoral head necrosis (FHN) occurs in fast growing birds such as broilers, and is characterised by the head of the femur separating from the bone shank. The femur head is often found in the acetabulum of the hip during processing. FHN most often occurs in 3–4 week old birds, although problems

are sometimes seen as early as 2 weeks. The affected bone is often ochre coloured and porous, and its brittle nature leads to the alternate name of brittle bone disease. As much as 30–40% of a flock can be affected to some degree although, for reasons unknown, the condition occurs sporadically in certain locations. Treatment and prevention relate to optimising the bird's nutrient intake (vitamins), although this action is not always successful. The condition is sometimes associated with enteritis and malabsorption so the vitamin status of these birds is often suspect. Once lameness and reluctance to move is noted, treatment with water soluble vitamins may help to prevent increased severity of FHN but complete recovery is seldom observed.

Foot pad dermatitis

Lesions to the footpads of birds (bumble foot) cause problems with locomotion and provide a route for infection by bacteria. It arises mainly as a result of poor litter conditions, whether nutritionally induced or otherwise. Birds fed biotin-deficient diets show a characteristic foot pad dermatitis that is responsive to increased biotin levels in the diet. Caged birds show a higher incidence than floor-managed birds so physical abrasion to the footpad would also seem to be a factor.

Rickets

Rickets occurs mostly in young birds, the main characteristic being inadequate bone mineralisation. They exhibit lameness, usually around 10–14 days of age. Their bone and beak become rubbery and the rib cage is flattened and beaded at the attachment to the vertebrae. Rickets is not caused by a failure in the initiation of bone mineralisation but rather in the early maturation of this process. Calcium deficiency at the cellular level is the main problem although this can be induced by feeding diets deficient

Measuring performance



Fact

Feed problems may be dramatic but typically they are insidious and difficult to identify

Lack of performance can be explained by: the bird itself, the environment (management), and health and/or nutrition. Although each class of bird has unique parameters that need to be assessed, in terms of nutrition the important matter is the measurement of feed quality and consistency.

If a claim for damages arises there are two clearly defined conditions that must be met: first, it must be shown that the feed supplied to the farm was in some way defective, and that a potential feed problem was the most likely cause for loss of performance and, secondly, it is the farmer's task to quantify the losses incurred in a transparent and verifiable way.

Feed impact on performance

A number of feed-related factors have a direct impact on bird performance, some of which can be controlled if sensible measures are taken:

Feed disorders

There are very few feed (nutritional) disorders that will cause an acute loss in production and/or an increase in mortality: generally, a deficiency is characterised by a general and gradual tail off in performance. If feed (or water) is absent entirely, the performance will drop sharply and immediately: this is most noticeable in laying hens. Surprisingly, production can return to normal levels fairly quickly. Always look for patterns when considering feed problems: did the problem coincide with a feed delivery and are other house/farms similarly affected? Any form of nutrient deficiency will result in a production slump because a deficiency of any nutrient cannot be tolerated for any length of time. The production curve shown in Figure 18.1 illustrates typical deviations seen in laying flocks.

As far as broiler feed problems are concerned the situation is more complex.

A gross error in the feed (10 times the ionophore level, for example) will cause an immediate drop in feed intake and growth, and an increase in mortality. The latter are usually easy to diagnose and remedy but the loss of production is more of a problem. From the response data for both protein and energy carried in Chapter 10, it can be seen that relatively large differences in feed specifications cause relatively small differences in performance – so it not always possible to know if any production drops seen are because of the feed, or some other factor. Some factors which will cause immediate production problems are:

- **Salt:** if high levels of salt are included in the diet there will be an increase in water consumption and the litter will become wet and slimy. The secondary effects of high salt intake can cause mortality and reach alarming proportions in a short space of time. When the feed is corrected, the mortality usually stops but growth and production do not return to their normal levels. When salt levels are too low, other symptoms develop: the flock becomes nervous (agitation and a distressed vocalisation), scratching as if looking for something, pecking is increased, and feathers are found in the digestive tract and the house.
- **Limestone:** if the limestone (calcium) is left out of a layer or broiler diet there is a dramatic increase in small, soft shelled eggs and after a few days, production drops sharply. There may also be increased mortality from calcium depletion (cage layer fatigue). By hand feeding oyster shell or limestone grit, the problem can be rectified reasonably quickly. In broiler rickets, soft keel bones and rubbery beaks may be seen: in this instance too much calcium is as likely to be

the problem as too little (see Chapter 12).

- **Phosphorus:** too little phosphorus in the diet causes a dramatic drop in feed intake. A surplus of calcium may complicate the issue. Ensure that the correct Ca:P ratio is achieved (Chapter 10).
- **Protein:** if protein levels are too low (only grain fed for, example), there will be an increase in nervousness, peck outs, poor albumin quality and low protein level from feed analysis. In broilers there may be ruffled and broken feathers: primary feathers on the wings stick out in a characteristic manner, which is why they are sometimes termed helicopter chicks.
- **Fat:** persistently low fat levels (not in the short term) result in low body weight gains, a drop in egg size and low fat in feed analysis. Rancid fat, on the other hand, causes immediate and dramatic feed refusal – one of the few things that has this effect.
- **Additives:** almost all of the feed additives used in the feed mill will cause irretrievable damage if fed at high levels. Usually, high levels of feed additives cause feed refusal, and birds show similar signs to those observed with a sodium or protein deficiency as feed intake drops. Nicarbazin causes shell-less eggs, loss of pigment of brown eggs, lowered hatch of fertile eggs, and a positive assay of feed for nicarbazin. Monensin causes reduced feed consumption, the birds lack co-ordination and show signs of paralysis and a positive feed assay for monensin.
- **Feed Ingredients:** substandard fishmeal or other protein sources can cause a fairly dramatic drop in growth and or production but

Appendix



Ingredient quality checklist

Physical quality control checklist for incoming raw materials

Raw material:

Date: / /

	Yes	No	Comments
Visual appearance			
Off colour			
Insect manifestation			
Foreign materials			
Texture deviation			
Flow characteristics			
Lumps			
Grind			
Grade of maize			
Taste			
Rancid			
Mouldy			
Burnt			
Off taste			
Smell			
Rancid			
Mouldy			
Burnt			
Off smell			
Feel			
Wet			
Temperature build up			
Bulk density			

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